

Project Design Phase-II

S.No	Component	Description	Technology
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1	User Interface	Transaction input, monitoring dashboard, fraud result display	HTML, CSS, JavaScript / React (Optional)
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Technology Stack (Architecture & Stack)

Date	12 February 2026
Team ID	LTVIP2026TMIDS64949
Project Name	Online Payments Fraud Detection using Machine Learning
Maximum Marks	4 Marks

Technical Architecture

2	Application Logic	API endpoints, transaction routing,	Python (Flask / FastAPI)
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		validation	
3	Preprocessing Pipeline	Missing value handling, encoding, scaling, feature engineering	scikit-learn, pandas, NumPy
4	Machine Learning Model	Fraud detection & probability scoring	Random Forest / XGBoost / Logistic Regression
5	Database / Storage	Transaction records, logs, model artifacts	MySQL / PostgreSQL / MongoDB

2	Security Implementations	Secure transaction handling, encryption, input validation	HTTPS, JWT/Auth, Data Encryption
3	Scalable Architecture	Modular layered design; cloud-ready	3-tier architecture, Docker/Kubernetes (Future)
4	Availability	Continuous fraud monitoring support	Cloud VM, Load Balancer (Future)
5	Performance	Real-time lowlatency fraud detection	Optimized inference pipeline, caching (Future)